ollision close-up - longest sequence, 6 contacts from t = 0.4099 s, wall 1 (green / red dotted: contact made

Shuttle and tip relative to the contact boundary

x = gc: nominal contact (tip air gap = h_eff)

shuttle approach - gc
tip r*tau - gc

+ls = 25 nm: vent sealed (chi = 1)

Nominal overlap and the tip-vent sealing window

Gap closure: h(tip) = h_eff + softpos(gc - r*tau)

hd = 2Tp/ep = 75 nm: coating-equivalent gap

h_eff + ls = 30 nm: tip vent starts to seal

h_eff = 5 nm: residual gap (floor)

tip air gap (nm)

Bending of the compliant electrodes

first contact 134.7 us, peak overlap 10.5 nm

delta = 0

-ls: vent open (chi = 0)

Velocities normal to the wall

shuttle approach
tip r*taudot

shuttle in 1.3, out -1.08 mm/s: e = 0.835

Forces on the tip coordinate (+ = into the wall)

bending
wall contact
electrostatic
squeeze film
beam damping

uN

Output voltage

Vout (mV)

time from first nominal contact (ms)

Phase portrait at the contact boundary

r*taudot (mm/s)

delta = r*tau - gc (nm)